YEAR 7 · MATHS UNITS 4 AND 5 · EXTRA HOMEWORK 12

What's Left of 180°?

Add and take away · decimals · a walk and a shop · angles on a straight line

Name: _____ **Class:** _____

Date given: _____ **Date due:** Monday 16 November 2026

Why this sheet exists

Every question here is an **add** or a **take away** — the angles too. An angle on a straight line is what is left of **180°**.

The steps are printed as **boxes**. Fill the boxes — that *is* the method. An **orange box** means: write a **0** here.

No calculator. No protractor. Write the **calculation** first, then the answer.

Part 1 · Add and take away

X1 Make the total

Write the missing number. (a) is done for you.

(a) 7 + 3 = 10 **(b)** 60 + _____ = 100 **(c)** 35 + _____ = 100

(d) 0.4 + _____ = 1 **(e)** 0.75 + _____ = 1 **(f)** 100 + _____ = 180

(g) 130 + _____ = 180 **(h)** 95 + _____ = 180 **(i)** 270 + _____ = 360

X2 In your head

No column grid this time. (a) is done for you.

(a) 0.5 + 0.5 = 1 **(b)** 1.2 + 0.8 = _____ **(c)** 2.5 + 2.5 = _____

(d) 4.6 + 0.4 = _____ **(e)** 0.9 + 0.2 = _____ **(f)** 3 − 0.5 = _____

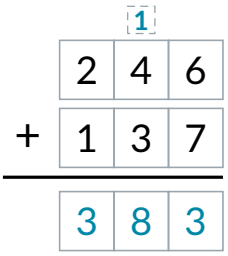
(g) 10 − 0.1 = _____ **(h)** 1 − 0.25 = _____ **(i)** 7.5 − 2.5 = _____

(j) 6 − 1.5 = _____ **(k)** 180 − 90 = _____ **(l)** 360 − 180 = _____

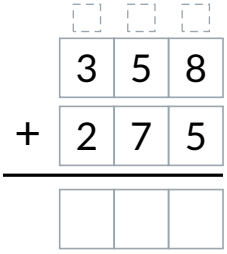
X3 Add in columns

A carried 1 goes in the small dashed box. In (c), write the numbers in the grid yourself.

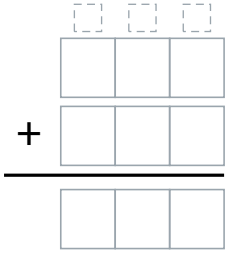
(a) 246 + 137 done



(b) 358 + 275



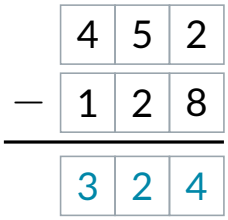
(c) 409 + 96



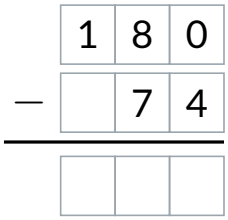
X4 Take away in columns

The first number goes on top. In (c), write the numbers in the grid yourself.

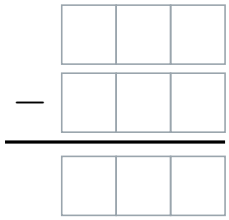
(a) 452 − 128 done



(b) 180 − 74



(c) 360 − 145

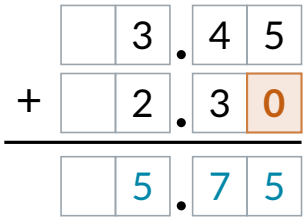


X5 Add decimals

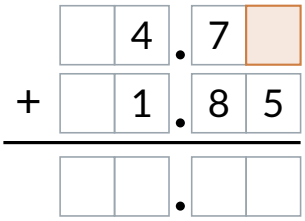
Remember from Unit 4

Line up the points. The point is printed in every row — it cannot move.
An empty place gets a **0**: 2.3 = 2.30 and 8 = 8.00.

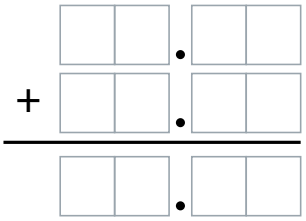
(a) 3.45 + 2.3 done



(b) 4.7 + 1.85



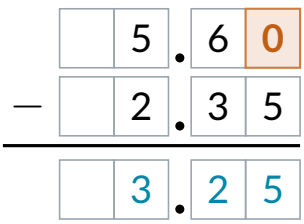
(c) 12.6 + 0.75



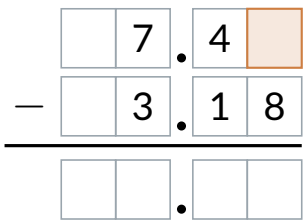
X6 Take away decimals

In (d), write the numbers in the grid yourself.

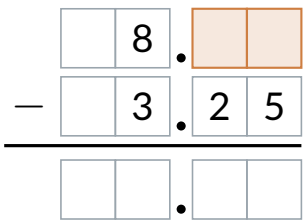
(a) 5.6 − 2.35 done



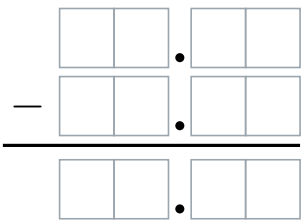
(b) 7.4 − 3.18



(c) 8 − 3.25



(d) 10 − 4.6



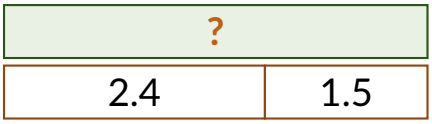
X7 Bar models

How a bar model works

The long bar on top is the **whole**. The two bars under it are the **parts**.
Whole missing? **Add** the parts. A part missing? **Take away**: whole − part.

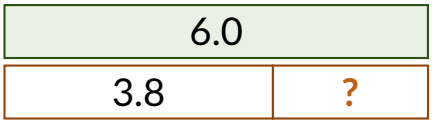
Find each **?**. (a) is done for you.

(a)



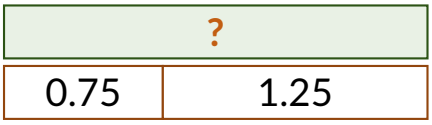
Calculation: 2.4 + 1.5 = 3.9 **?** = **3.9**

(b)



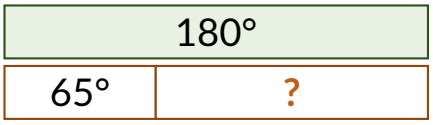
Calculation: _____ **?** = _____

(c)



Calculation: _____ **?** = _____

(d)

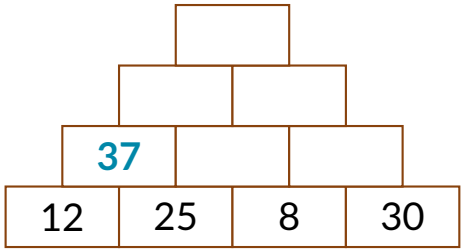


Calculation: _____ **?** = _____

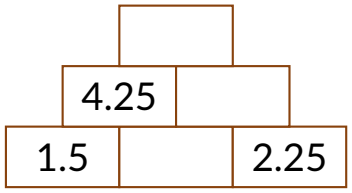
X8 Number walls

Each brick is the **two bricks under it, added**. Fill every empty brick.

(a)



(b)



A bottom brick is missing: 1.5 + ? = 4.25.
Take away.

X9 Add or take away?

Word help — which one?

add total · altogether · the sum of · more (it grows)
take away the difference between · how much more · change · left · subtract
Careful: *subtract 2 from 9* means 9 − 2. The second number comes first.

Write the **calculation first**. Then write the answer.

(a) What is the total of 2.5 and 1.25?

Calculation: _____ **Answer:** _____

(b) What is the difference between 9.6 and 4.2?

Calculation: _____ **Answer:** _____

(c) A gecko is 4.5 cm long. It grows 1.8 cm. How long is it now?

Calculation: _____ **Answer:** _____

(d) A bottle holds 1.5 litres of water. Mr Bowen drinks 0.35 litres. How much is left?

Calculation: _____ **Answer:** _____

(e) Mr Bowen walks 2.4 km on Monday and 3.15 km on Tuesday. How far does he walk altogether?

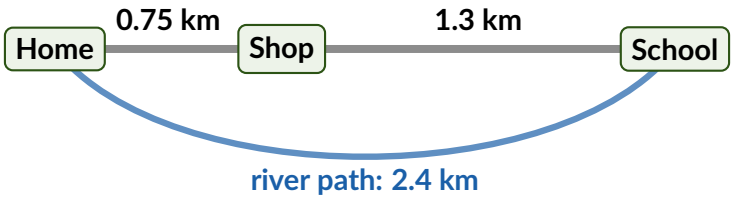
Calculation: _____ **Answer:** _____

(f) Subtract 2.7 from 6.

Calculation: _____ **Answer:** _____

X10 Mr Bowen's walk to school

Mr Bowen walks to school. He can go past the shop, or along the river.



(a) How far is it from home to school, past the shop?

Calculation: _____ **Answer:** _____

(b) How much longer is the river path?

Calculation: _____ **Answer:** _____

(c) He walks to school and home again, past the shop both times. How far does he walk altogether?

Calculation: _____ **Answer:** _____

(d) One day he gets to the shop and remembers his laptop is at home. He walks home, back to the shop, and on to school. How far does he walk that morning?

Calculation: _____ **Answer:** _____

X11 Mr Bowen goes shopping

School shop	
notebook	2.50 dollars
pen	1.25 dollars
ruler	0.80 dollars
eraser	0.45 dollars
glue	1.60 dollars

change: the money you get back. You pay 5 dollars, it costs 3 dollars: your change is $5 - 3 = 2$ dollars.

(a) Mr Bowen buys a pen and a ruler. What is the total?

Calculation: _____ **Answer:** _____

(b) He buys a notebook. He pays with 5 dollars. What is his change?

Calculation: _____ **Answer:** _____

(c) How much more is the glue than the ruler?

Calculation: _____ **Answer:** _____

(d) He buys a notebook, a pen and an eraser. He pays with 5 dollars. What is his change? Two steps:

Step 1 add: _____ + _____ + _____ = _____

Step 2 take away: 5.00 − _____ = _____ dollars

X12 Mr Bowen's homework

Circle **right** or **wrong**. If it is wrong, write the correct answer. Check with a column grid if you need to.

(a) 4.5 + 2.35 = 2.80 right · wrong correct: _____

(b) 6 − 2.4 = 4.4 right · wrong correct: _____

(c) 3.7 + 1.5 = 5.2 right · wrong correct: _____

(d) 9 − 3.25 = 6.25 right · wrong correct: _____

Choose **one** wrong answer. What did Mr Bowen do wrong?

Start: **"In ( ), Mr Bowen …"**

X13 Now you write the question

Write a word problem for 5.00 − 3.45 = 1.55

Your word problem must have: tick each one

- ☐ Mr Bowen and a real thing to buy
- ☐ the word **change**
- ☐ dollars
- ☐ full sentences
- ☐ a question at the end
- ☐ the answer 1.55 dollars

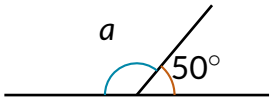
Part 2 · Angles: what is left?

X14 What is left of 180°?

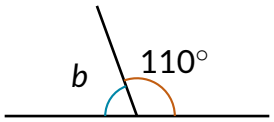
Remember from 5.1

Angles on a **straight line** add up to **180°**. A small square means a **right angle**: **90°**.
Angles **around a point** add up to **360°**.

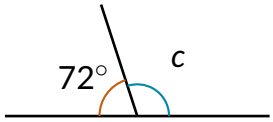
Find each letter. Take the angle away from 180°. (a) is done for you.



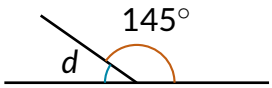
(a) 180 − 50 = **130°**



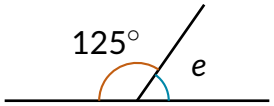
(b) 180 − _____ = _____



(c) 180 − _____ = _____



(d) 180 − _____ = _____



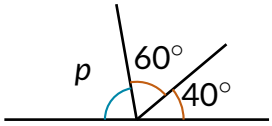
(e) _____ − _____ = _____



(f) _____

X15 Two angles, then what is left

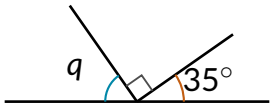
Step 1: **add** the two angles. Step 2: **take away** from 180°. (a) is done for you.



1. 40 + 60 = 100

2. 180 − 100 = 80

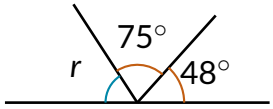
$p = 80°$



1. _____

2. _____

q = _____



1. _____

2. _____

r = _____

X16 Straight line or not?

Two angles sit side by side. Add them. Do they make a **straight line**? Row 1 is done for you.

The two angles	Add them	Straight line?
95° and 85°	95 + 85 = 180°	yes
72° and 106°		yes · no
129° and 51°		yes · no
88.5° and 91.5°		yes · no
64° and 117°		yes · no

X17 Angles in words

Word help

straight line	180°	**full turn, around a point**	360°
right angle	90°	**equal**	the same size

You can draw a sketch. Write the **calculation first**.

(a) Two angles make a straight line. One of them is 70°. What is the other one?

Calculation: _____ **Answer:** _____

(b) Two angles make a straight line. They are equal. How big is each one?

Calculation: _____ **Answer:** _____

(c) Three angles make a straight line. Two of them are 50° and 60°. What is the third?

Calculation: _____ **Answer:** _____

(d) A right angle and one more angle make a straight line. How big is the other angle?

Calculation: _____ **Answer:** _____

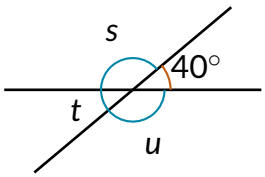
(e) Two angles make a full turn. One of them is 200°. What is the other?

Calculation: _____ **Answer:** _____

X18 Two lines cross

Remember from 5.2

When two lines cross, **opposite** angles are **equal**. Angles **next to** each other make a straight line: **180°**.

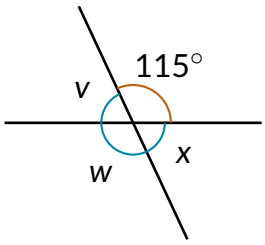


(a)

t = 40°

s = _____

u = _____



(b)

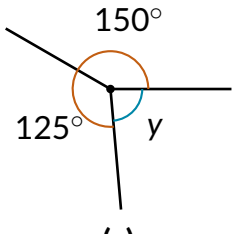
v = _____

w = _____

x = _____

X19 Around a point

Step 1: add the angles you know. Step 2: take away from **360°**.

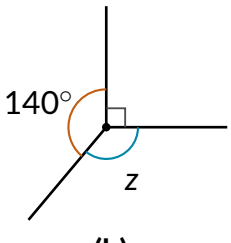


(a)

1. _____

2. _____

y = _____



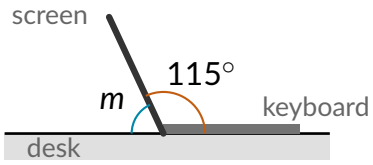
(b)

1. _____

2. _____

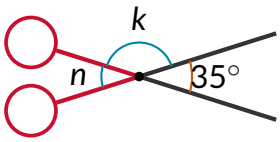
z = _____

X20 Mr Bowen's angles



(a) Mr Bowen opens his laptop. The screen makes 115° with the keyboard. The desk is a straight line. Find *m*.

Calculation: _____ *m* = _____



(b) Mr Bowen's scissors are two straight lines that cross. The blades are open at 35°.

n = _____ because _____

k = _____ because _____

X21 Mr Bowen's angle homework

Circle **right** or **wrong**. Write the calculation that shows it.

(a) A right angle and 80° make a straight line. right · wrong _____

(b) Two lines cross. One angle is 50°, so the angle **opposite** it is 130°. right · wrong _____

(c) Three angles around a point are 100°, 120° and 140°. right · wrong _____

(d) On a straight line, one angle is 75°, so the other is 115°. right · wrong _____

X22 Now you write the question

Write a word problem for $180° - 65° = 115°$

Your word problem must have: tick each one

- ☐ a real thing (a road, a ladder, a door …)
- ☐ the words **straight line**
- ☐ full sentences
- ☐ Mr Bowen
- ☐ a question at the end
- ☐ the answer 115°

Answer Key — Teacher Copy

X1 Make the total

(b) 40 (c) 65 (d) 0.6 (e) 0.25 (f) 80 (g) 50 (h) 85 (i) 90

(e) Expect 0.35 ("75 and 35 make 100" — no: 75 + 25) or 0.25 written as 25. (f)–(i) are the angle facts in disguise: the student who finds them easy already has Part 2.

X2 In your head

(b) 2 (c) 5 (d) 5 (e) 1.1 (f) 2.5 (g) 9.9 (h) 0.75 (i) 5 (j) 4.5 (k) 90 (l) 180

(e) Expect 0.11 ("9 and 2 make 11"): ten tenths make 1. (g) Expect 9.1 or 10.9. (b), (c), (d), (i) are whole numbers on purpose — a student who writes 2.0 or 5.0 is right.

X3 Add in columns

(b) 358 + 275 = **633** (carries in the tens and the hundreds) (c) 409 + 96 = **505**

(c) is the alignment check. Writing 96 under the 40 of 409 gives 1369; ones go under ones.

X4 Take away in columns

(b) 180 − 74 = **106** (c) 360 − 145 = **215**

Both have a 0 in the ones on top. The classic slip is 0 − 4 = 4 (take the small from the big, whichever row it is in): 180 − 74 then comes out as 114, and 360 − 145 as 225. Ask: "Can you take 4 from 0?" — then exchange a ten.

X5 Add decimals

(b) 4.70 + 1.85 = **6.55** (c) 12.60 + 0.75 = **13.35**

(c) is empty on purpose: 12.6 needs the tens box, and 0.75's 0 goes in the ones. A student who writes 12.6 starting in the first box has lined up the left edge, not the point — the point is printed, so ask where it is.

X6 Take away decimals

(b) 7.40 − 3.18 = **4.22** (c) 8.00 − 3.25 = **4.75** (d) 10.0 − 4.6 = **5.4**

(c) is the one to mark carefully. 5.25 means 8 − 3 and the .25 copied down; the orange boxes are there to stop exactly that. (b): 4.38 is "8 − 0, take the smaller from the bigger". (d) needs 10 in the tens and ones boxes.

X7 Bar models

(b) 6.0 − 3.8 = **2.2** (c) 0.75 + 1.25 = **2** (2.00) (d) 180 − 65 = **115**°

The picture chooses the operation: whole missing, add; part missing, take away. Expect 6.0 + 3.8 = 9.8 in (b) from a student adding every two numbers — point at the top bar: 9.8 is bigger than the whole. (d) is Part 2 in a bar.

X8 Number walls

(a) middle row 37, 33, 38; then 70, 71; top **141**.
(b) missing brick 4.25 − 1.5 = **2.75**; then 2.75 + 2.25 = 5.00; top 4.25 + 5.00 = **9.25**.

(b) is the only take-away: a student who adds 1.5 + 4.25 for the missing brick (5.75) gets a wall that does not add up — ask them to check the brick above it.

X9 Add or take away?

(a) 2.5 + 1.25 = 3.75 (b) 9.6 − 4.2 = 5.4 (c) 4.5 + 1.8 = 6.3 cm (d) 1.5 − 0.35 = 1.15 litres
(e) 2.4 + 3.15 = 5.55 km (f) 6 − 2.7 = 3.3

Mark the calculation first. (d) Expect 1.25: in 1.50 − 0.35 the student wrote 0 − 5 as 5 instead of exchanging. (f) is the trap: 2.7 − 6 is the English error, not an arithmetic one — have them read the Careful line aloud. (c) is HW 12's gecko, still growing.

X10 Mr Bowen's walk to school

(a) 0.75 + 1.3 = **2.05** km (b) 2.4 − 2.05 = **0.35** km (c) 2.05 + 2.05 = **4.1** km
(d) 0.75 + 0.75 + 0.75 + 1.3 = **3.55** km

(a) Expect 0.88 from a student who did not line up the points: 1.3 written under the 75 as if it were 0.13 (1.3 is 1.30). (b) needs (a) first: a student who writes 2.4 − 1.3 or 2.4 − 0.75 compared the river with one part of the road. (d) is three walks of 0.75 km, not two: home to shop, shop to home, home to shop again.

X11 Mr Bowen goes shopping

(a) 1.25 + 0.80 = 2.05 dollars (b) 5.00 − 2.50 = 2.50 dollars (c) 1.60 − 0.80 = 0.80 dollars
(d) Step 1: 2.50 + 1.25 + 0.45 = 4.20; Step 2: 5.00 − 4.20 = **0.80** dollars

(c) "How much more" is a take-away; expect 2.40 from a student who added. (d) is two steps, the shopping version of X15: a student who stops at 4.20 has answered "how much did it cost", not "what is his change".

X12 Mr Bowen's homework

(a) wrong, **6.85** (b) wrong, **3.6** (c) **right** (d) wrong, **5.75**

(a) he lined up the right-hand edges, not the points (45 + 235 = 280). (b) he took 2 from 6 and copied the .4 down. (d) the same error as (b) and X6(c): he took 3 from 9 and kept the .25. (c) is right so the answer is not "all wrong". For the sentence, accept any one error described in words: *"In (b), Mr Bowen took 2 from 6 and did not take away the 0.4."*

X13 Now you write the question

Check, in order: something Mr Bowen buys for 3.45 dollars; he pays 5 dollars; it asks for the change; a question mark; full sentences.
Model: *"Mr Bowen buys a sandwich for 3.45 dollars. He pays with 5 dollars. How much change does he get?"*
The usual failure: a question whose answer is 3.45 ("how much is the sandwich?") or 8.45 (it adds). Ask the student to answer their own question.

X14 What is left of 180°?

(b) 180 − 110 = **70**° (c) 180 − 72 = **108**° (d) 180 − 145 = **35**° (e) 180 − 125 = **55**° (f) 180 − 147.5 = **32.5**°

(c) and (e) have the letter on the right. Expect 252 for (c) from a student who adds. (f) is X6's skill: 180.0 − 147.5, a 0 after the point.

X15 Two angles, then what is left

(b) 35 + 90 = 125, 180 − 125 = **55**° (c) 48 + 75 = 123, 180 − 123 = **57**°

(b) has no number for the right angle: expect 180 − 35 = 145 from anyone who skipped the square.

X16 Straight line or not?

72 + 106 = 178, **no** 129 + 51 = 180, **yes** 88.5 + 91.5 = 180, **yes** 64 + 117 = 181, **no**

178 and 181 are the point: "nearly 180" is not a straight line. A "yes" for either means the student looked instead of adding.

X17 Angles in words

(a) 180 − 70 = **110°** (b) 180 ÷ 2 = **90°** (c) 50 + 60 = 110, 180 − 110 = **70°** (d) 180 − 90 = **90°** (e) 360 − 200 = **160°**

Every one is decided by its English. (b) "equal" is the only divide; expect 180 from a student who missed it. (d) "right angle" is the 90 they must supply. (e) "full turn" is 360: expect 180 − 200 (impossible) from a student who uses 180 for everything.

X18 Two lines cross

(a) t = 40° (opposite), s = u = 140° (180 − 40) (b) w = 115° (opposite), v = x = 65° (180 − 115)

(b) gives only one angle. A student who writes 115 for all three has called every angle "opposite"; v is *next to* 115°.

X19 Around a point

(a) 150 + 125 = 275, 360 − 275 = **85°** (b) 90 + 140 = 230, 360 − 230 = **130°**

Expect 180 − 275 (impossible) from a student who uses 180 everywhere: ask "line or point?". (b) has a square for the 90.

X20 Mr Bowen's angles

(a) 180 − 115 = **65°** (b) n = **35°** because opposite angles are equal; k = **145°** because next-door angles make 180° (180 − 35).

(a) is X14 with a picture. (b): n is between the handles, k between a blade and a handle. Accept the reason in the student's own words.

X21 Mr Bowen's angle homework

(a) **wrong**: 90 + 80 = 170, not 180 (b) **wrong**: opposite angles are equal, so it is 50° (130° is next to it)
(c) **right**: 100 + 120 + 140 = 360 (d) **wrong**: 180 − 75 = 105, not 115

Three different errors on purpose: (a) forgets that a right angle is a number (90), (b) swaps opposite and next-door, (d) is an arithmetic slip that "looks about right". A student who marks (c) wrong probably added to 180: "around a point" is 360.

X22 Now you write the question

Check, in order: a straight line (named); one angle of 65°; it asks for the other angle; a question mark; full sentences.
Model: *"Mr Bowen leans a ladder on a wall. The ladder makes 65° with the ground. The ground is a straight line. What is the angle on the other side of the ladder?"*
The usual failure: a question whose answer is 65° (it gives 115 and asks for 65 — still a good question, but not this one) or one about a point (360).